EMU414 - Mathematical Programming and Computer Applications
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Homework 2
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Homework 2

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Questions

Question 1:
The US Forest Service requires an allocation model for managing the 788-thousand-acre Wagonho National Forest. This allocation model will distribute forest areas according to timber, grazing, recreation, environmental, national conservation, and other needs. First, the entire area is divided into 7 analysis areas. Then, 3 management policies, called prescriptions, are determined for each area. The optimization model needs to allocate the analysis areas to these three prescriptions. The allocation plan must yield at least 40 million board feet of timber, at least 5 animal unit months (AUM) of grazing capacity, and an average wilderness index of at least 70. Let us develop a mathematical optimization model for this allocation plan. The input data and mathematical model for the problem is available in our lecture notes.

Mathematical Notation

Table 1: Sets and Indices

Symbol	Description
$i \in I$	Analysis areas, $I = \{1, 2, 3, 4, 5, 6, 7\}$
$j \in J$	Prescriptions, $J = \{1, 2, 3\}$

Table 2: Parameters

Symbol	Description
a_{ij}	Total acres in analysis area i under prescription j
A_i	Total acres available in analysis area i
t_{ij}	Timber yield (board feet per acre) for area i, prescription j
g_{ij}	Grazing capacity (AUM per acre) for area i, prescription j
w_{ij}	Wilderness index for area i, prescription j
n_{ij}	Net value per acre for area i, prescription j

Table 3: Decision Variables

Symbol	Description
x_{ij}	Acres allocated to analysis area i under prescription j

Mathematical Model

$$\max \quad \sum_{i \in I} \sum_{j \in J} n_{ij} \cdot x_{ij} \tag{1}$$

$$\text{s.t.} \quad \sum_{j \in J} x_{ij} = A_i \quad \forall i \in I \tag{2}$$

$$\sum_{i \in I} \sum_{j \in J} t_{ij} \cdot x_{ij} \geq 40000 \tag{3}$$

$$\sum_{i \in I} \sum_{j \in J} g_{ij} \cdot x_{ij} \geq 5 \tag{4}$$

$$\sum_{i \in I} \sum_{j \in J} w_{ij} \cdot x_{ij} \geq 70 \cdot \sum_{i \in I} A_i \tag{5}$$

$$x_{ij} \geq 0 \quad \forall i \in I, \forall j \in J \tag{6}$$

Model Explanation

The objective function (1) maximizes the total net value obtained from the forest allocation. Constraint (2) ensures that the total acres allocated across all prescriptions equals the available acres for each analysis area. Constraint (3) guarantees that timber production meets the minimum requirement of 40,000 board feet. Constraint (4) ensures grazing capacity is at least 5 AUM. Constraint (5) maintains the average wilderness index at or above 70. Finally, constraint (6) enforces non-negativity of decision variables.

Gurobi Results

The optimal solution obtained from Gurobi is as follows:

Table 4: Optimal Decision Variable Values

Analysis Area	Prescription	Acres Allocated
1	3	75.00
2	1	90.00
3	1	140.00
4	3	60.00
5	2	154.00
5	3	58.00
6	3	98.00
7	3	113.00

Objective Function Value: 322,515

Question 2

An operations research (OR) specialist has started a project consisting of 7 tasks. Some tasks can be done independently of others, but some tasks have predecessors. For example, to solve a model with Gurobi, the model must first be formulated. If formulating the model takes 4 days, the Gurobi phase can only begin 4 days after the formulation starts.

The OR specialist has determined the minimum and maximum time to complete each task. They need to choose the time they will allocate to each task within these bounds. The OR specialist

has also calculated a cost related to time for each task. Completing tasks in a short time is more expensive than spreading them over a longer period. The table below shows the minimum and maximum time (in days) that can be allocated to each task and the associated costs. For example, if 6 days are allocated to the first task, the cost is 1600, while if 12 days are allocated, the cost is 1000. The cost-time relationship between the minimum and maximum values changes linearly. This way, the cost for any given duration can be calculated.

The OR specialist wants to complete all work within 40 days at minimum total cost, and is looking for answers on when to start each task and how much time to allocate to each task. They have gathered their data in Table 5. Develop a linear programming (LP) model that plans the time-cost schedule for this seven-task project and minimizes the total cost. The model aims to find the start times of tasks and the durations allocated to each task. Solve it using the Gurobi solver. Report the decision variable values and objective function for the optimal solution. (Verification hint: in the optimal solution, the start time of task 3 is 8, and the duration allocated to task 3 is 16.)

Table 5: Question 2 data

Task No	Min. Time	Max. Time	Min. Cost	Max. Cost	Predecessor
1	6	12	1600	1000	none
2	8	16	2400	1800	none
3	16	24	2900	2000	2
4	14	20	1900	1300	1, 2
5	4	16	3800	2000	3
6	12	16	2900	2200	3
7	2	12	1300	800	4

Mathematical Notation

Table 6: Sets and Indices

Symbol	Description
$i \in T$	Tasks, $T = \{1, 2, 3, 4, 5, 6, 7\}$

Table 7: Parameters

Symbol	Description
d_i^{min}	Minimum duration for task i (days)
d_i^{max}	Maximum duration for task i (days)
c_i^{min}	Cost at minimum duration for task i
c_i^{max}	Cost at maximum duration for task i
α_i	Cost slope for task i: $\alpha_i = \frac{c_i^{max} - c_i^{min}}{d_i^{max} - d_i^{min}}$
P_i	Set of predecessor tasks for task i
D	Project deadline (40 days)

Table 8: Decision Variables

Symbol	Description
s_i	Start time of task i (days)
d_i	Duration allocated to task i (days)

Mathematical Model

$$\min \quad \sum_{i \in T} \left[c_i^{min} + \alpha_i \cdot (d_i - d_i^{min}) \right] \tag{7}$$

$$\text{s.t.} \quad d_i^{min} \leq d_i \leq d_i^{max} \quad \forall i \in T \tag{8}$$

$$s_i \geq s_p + d_p \quad \forall i \in T, \forall p \in P_i \tag{9}$$

$$s_i + d_i \leq D \quad \forall i \in T \tag{10}$$

$$s_i \geq 0 \quad \forall i \in T \tag{11}$$

Model Explanation

The objective function (7) minimizes the total cost across all tasks. Since completing a task faster is more expensive, the cost decreases linearly as the duration increases. Constraint (8) ensures each task's duration stays within its allowed bounds. Constraint (9) enforces precedence relationships: a task cannot start until all its predecessors are completed. Constraint (10) guarantees all tasks finish before the project deadline of 40 days. Constraint (11) ensures non-negative start times.

Gurobi Results

The optimal solution obtained from Gurobi is as follows:

Table 9: Optimal Schedule

Task	Start Time	Duration	Cost
1	0.00	12.00	1000.00
2	0.00	8.00	2400.00
3	8.00	16.00	2900.00
4	12.00	20.00	1300.00
5	24.00	16.00	2000.00
6	24.00	16.00	2200.00
7	32.00	8.00	1000.00

Objective Function Value: 12,800